AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2024

Computer Science Paper II

Time: 1 hour 30 minutes Marks: 35

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are EIGHT questions. Answer ALL questions. Question 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 3 Marks)

Sara is a system administrator responsible for managing multiple servers in a data centre. She is considering using a Command Line Interface (CLI) to perform the tasks.

Give THREE advantages of using CLI in the given scenario.

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_____

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Q.2. (Total 3 Marks)

Aliza is a software quality assurance analyst responsible for testing a new software application. She wants to know the phases of the Software Development Life Cycle (SDLC) that directly impact her testing efforts.

Why are the given phases of SDLC MORE relevant to Aliza's role?

Phase	Reason
Analysis	
Design	
Testing	

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Q.3. (Total 5 Marks)

Give suitable data type that can be used in C++ programming for the given situations.

Situation	Data Type
In a user registration system, you need to store the gender of each user such as 'M' for male or 'F' for female.	
You are developing a program to input and display the names of students.	
In a voting application, you need to store the age of eligible voters.	
You are creating a weather monitoring program that records temperature readings.	
In an inventory management system, you need to store prices of products.	

Q.4. (Total 3 Marks)

Compare for loop and while loop on the basis of the following features.

Feature	For Loop	While Loop
Initialisation		
Control		
Use case		

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Q.5. (Total 4 Marks)

Describe any FOUR advantages of using method overloading in comparison to method overriding.

Q.6. (Total 3 Marks)

i. Describe the primary purpose of using 'ifstream' mode. (1 Mark)

ii. Highlight TWO situations where you would need to use both 'ifstream' and 'ofstream' modes in the same program to read and write to a file. (2 Marks)

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Q.7. (Total 7 Marks)

EITHER

a. Write a C++ program that

- takes a positive integer ‘n’ as an input.
- print the sum of the ‘n’ even numbers.

OR

b. Write a C++ program that creates a simple ATM menu system which allows users to perform the following basic banking operations by using switch case statement.

- checking their balance
- making withdrawals
- making deposits
- exit

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Q.8. (Total 7 Marks)

EITHER

a. Write a C++ program that takes two strings as an input and

- print the length of both strings.
- check if both strings are identical.

OR

b. Write a C++ program using two user defined functions that

- takes length and width as an input.
- prints the area and perimeter of a rectangle.

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